

BIOLOGY BOARD QUESTION PAPER : FEBRUARY 2025**Time: 3 Hrs.****Max. Marks: 70****General Instructions:**

(1) The question paper is divided into four sections.

Section A: Q.1 contains ten multiple choice type of questions carrying one mark each.

Evaluation will be done for the first attempt only.

Section A: Q.2 contains eight very short answer type of questions carrying one mark each.

(2) Section B: Q.3 to Q.10 are short answer type questions carrying two marks each.

(Attempt any Eight)

(3) Section C: Q.11 to Q.26 are long answer type of questions carrying three marks each.

(Attempt any Eight)

(4) Section D: Q.27 to Q.31 are long answer type of questions carrying four marks each.

(Attempt any Three)

(5) Draw neat and labelled diagrams wherever necessary.

(6) Write the answers of each section on a new page.

SECTION – A

Q.1 Select and write the correct answer for the following multiple choice questions:

[10]

i. The aquatic plant _____ is known as "Terror of Bengal".

(A) Lotus (B) Pistia (C) Eichhornia (D) Vallisneria

ii. A flower shows dull colour, strong fragrance and edible pollen grains.

The type of pollination in such flowers is:

(A) Anemophily (B) Chiropterophilly (C) Ornithophilly (D) Entomophilly

iii. Part of the spore containing enzymes for penetration of corona radiata is _____.

(A) tail (B) middle part (C) head (D) neck

iv. In bird population of newly formed volcanic island, a rare allele of a gene was noted to be much more prevalent than the populations of the same species on the mainland. This is known as:

(A) Founder effect (B) Bottleneck effect (C) Bottle neck (D) Haldane effect

v. In a plant, deficiency symptom called "Die-back" of shoot was observed.

For more study material, question papers & updates — Scan the QR code or visit



This can be rectified by supplying _____.

- (A) iron (B) sulphur (C) zinc (D) copper

vi. Insect resistant property in cotton crop is developed by introducing a gene from _____.

- (A) *Bacillus thuringiensis* (B) *Bacillus subtilis*
(C) *Bacillus megaterium* (D) *Bacillus sphaericus*

vii. The type of interspecific interaction in which one species is harmed and the other is not affected is:

- (A) commensalism (B) parasitism (C) amensalism (D) mutualism

viii. Small pores present on woody stem surface with scar tissue which carry out gaseous exchange are:

- (A) stomata (B) hydathode (C) ventricle (D) lenticels

ix. In the given food chain, 1000 Joules of light energy falls on the producer.

How much energy is available for Lion?

Plants → Deer → Lion

(Producer) (Herbivore) (Carnivore)

- (A) 1 J (B) 10 J (C) 1000 J (D) 0.1 J

x. Glucagon is secreted by _____ cells of pancreas.

- (A) PP (B) alpha (C) delta (D) beta

Q.2 Answer the following questions:

[8]

i. What is the effect of excessive transpiration in leaves?

ii. Which method of plant breeding can be used to obtain many genetically identical saplings of endangered species?

iii. Define: Spermatogenesis.

iv. What is grand period of growth?

v. Write the anticodon of 5' AUG 3'.

vi. Mention pollination agents for maize flowers.

vii. Which is the first step in the process of absorption of water by roots from soil?

viii. Name the hormonal disorder caused due to hyperthyroidism in adults.



SECTION – B**Attempt any EIGHT of the following questions:****[16]**

- Q.3 Enlist any two domesticated species of honeybee and write their role in increasing crop yield.
- Q.4 i. Name the vein which carries oxygenated blood.
ii. How does portal vein differ from normal vein?
- Q.5 The health worker in hilly area will advise people to prevent which deficiency disease? Which micronutrient is deficient in the above area?
- Q.6 Name the vector responsible for causing malaria and filariasis respectively.
- Q.7 Name two chromosomal disorders which show presence of 47 chromosomes. Give their genotypes.
- Q.8 Which germinal layer produces the following structures?
i. Eustachian tube
ii. Pineal gland
iii. Mucous lining of alveoli
iv. Lining of middle ear
- Q.9 Due to environmental change in area, the fruits were taking more time for ripening. Which gaseous growth regulator can be suspected to induce fruit ripening? Give the source of the same in plants.
- Q.10 Explain two-celled stage of pollen grain in angiosperms.
- Q.11 What is the difference between Heterocatalytic and Autocatalytic function of DNA?
- Q.12 What is Transpiration?
How does it differ from guttation?
- Q.13 Give reason:
When Rh–ve mother conceives Rh+ve foetus, she is given injection of anti-Rh antibodies.
- Q.14 Identify the chromosomal aberrations:
i. $A \rightarrow BCDEF GH \rightarrow ABC | FGH$
ii. $ABCD \rightarrow ABCDB$
iii. $ABCD \rightarrow ABCDEF$
iv. $ACDEFG \rightarrow ADC | BEFG$



SECTION – C

Q.15 Complete the following table:

Organ Region Sympathetic effect Parasympathetic effect		
i. Heart beat	Increases	
ii. Blood vessels		Constricts
iii. Arterial B.P.		Decreases
iv. Pupil of eye		Constricts
v. Gastrointestinal movements	Retards peristalsis	
vi. Urinary bladder	Relaxes	

Q.16 i. Draw neat and labelled diagram to show T.S. of artery.

ii. Name two superficially placed arteries.

Q.17 Write any six characteristics of genetic code.

Q.18 i. In the process of DNA technology, if we suppose restriction enzymes are used to cut the

desired gene and donor DNA, then which problem will arise in formation of rDNA or chimeric DNA?

ii. Name any two restriction enzymes and their source organisms.

Q.19 i. Explain vestigial organs as evidence of evolution.

ii. Give any two examples of vestigial organs in human beings.

Q.20 What is water potential?

Write any four factors affecting water absorption in plants.

Q.21 Draw the developmental flow chart in plant from seed germination to death of plant.

Q.22 What is meant by biofertilizers?

Write the benefits of biofertilizer.

Q.23 What is biofortification? Explain with respect to GM food plants.

Q.24 Differentiate between habitat and niche.

Q.25 What is decomposition?

Give diagrammatic representation of "decomposition cycle."

Q.26 What is biomagnification?

Draw the flow chart and explain it with respect to DDT.



SECTION – D**Attempt any THREE of the following questions:****[12]**

Q.27 i. What is the location of blind spot in human eye?

ii. Write importance of retina in image formation.

iii. Name the cells which function in dim light and day light.

Q.28 With the help of a neat and labelled diagram, explain the conducting system of human heart.

Q.29 i. Define: Sphingomyelin

ii. Differentiate between chalazogamy and porogamy with respect to the entry of the pollen tube.

iii. Draw neat and labelled diagram to show mesogamy.

Q.30 i. Define: Infertility

ii. If a woman's fallopian tube is blocked, what treatment can be given to overcome infertility?

iii. What is sperm bank?

